

Scikit-Learn Complete Hand-Made Notes

1. What is Scikit-Learn?

- Python ML library for supervised and unsupervised learning

2. Main ML Categories

Supervised: Linear Regression, Logistic Regression, SVM

Unsupervised: K-Means, PCA

Ensemble: Random Forest, Gradient Boosting

3. ML Workflow

Data Load → Preprocess → Split → Train → Predict → Evaluate

4. Data Preprocessing

StandardScaler(), LabelEncoder(), OneHotEncoder(), SimpleImputer()

5. Algorithms

Regression: LinearRegression, DecisionTreeRegressor

Classification: LogisticRegression, SVC, KNN, GaussianNB

6. Metrics

Classification: Accuracy, Precision, Recall, F1

Regression: MSE, RMSE, MAE, R2

7. Train Test Split

train_test_split(X, y, test_size=0.2)

8. Cross Validation

cross_val_score(model, X, y, cv=5)

9. Hyperparameter Tuning

GridSearchCV(model, param_grid, cv=5)

10. Save & Load Model

pickle.dump(), pickle.load()

Revision:

Load → Clean → Split → Train → Predict → Evaluate → Tune → Deploy